Cleaner Air, Thinner Output? The Impact of EPA Greenbook Nonattainment Designations on the Economic Activities of Rust Belt Manufacturing Hubs

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1 Introduction

This paper studies how environmental regulations affect both economic activities and pollution by examining the consequences of an EPA nonattainment designation, which triggers stricter local compliance requirements. We focus our analysis on the industrial and manufacturing intensive Rust Belt region, and more specifically Wayne County, Michigan during the period between 2001 to 2010. Wayne County first appeared on the EPA's nonattainment list in 2005, which we take as the treatment year. Our identification strategy employs a two-way fixed effects model and synthetic controls where parallel trend assumptions may be violated. Overall, we find a statistically significant decline in overall GDP, but we do not find robust evidence that the decline is concentrated in a specific sector or that the designation leads to meaningful reductions in measured emissions over our sample period.

1.1 Motivation and Literature

First established in 1970, the US Clean Air Act is a comprehensive set of regulations aimed at addressing public health and welfare risks associated with emissions of hazardous air pollutants. Moreover, it grants the Environmental Protection Agency (EPA) authority to implement standards to monitor and regulate the emission of pollutants. Each year, the EPA Green Book provides county-level information on National Ambient Air Quality Standards (NAAQS) with the goal of providing a framework for an evidence-based regulation program advancing energy efficiency and tracking pollutants (U.S. Environmental Protection Agency).

Areas classified with the nonattainment status (indicating evidence of concentration of one or more pollutants exceeding the acceptable threshold set by the NAAQS) will be required to submit a state implementation plan (SIP) to the EPA, which outlines the state's approach to improving environmental standards as well as emissions data of relevant precursor pollutants (U.S. Environmental Protection Agency). Once approved by the EPA, states will work along with local authorities towards ensuring all establishments, such as industrial facilities and factories within the area, improve emission output by a federally mandated deadline. States must also demonstrate reasonable progress in the interim, as described in the 1990 Amendment to the Clean Air Act, and failure to do so may involve economic penalties levied against both the region and the facility. As of 2025, there are six active standards, including Ozone, Particulate Matter (PM 2.5), Sulphur Dioxide, Lead, Carbon Monoxide and Nitrogen Dioxide.

1.1.1 Literature Review

Using a theoretical framework, Porter and van der Linde (1995) hypothesize that stringent environmental regulations will ultimately stimulate innovation and enhance long-run market competition. In practice, since firms operate under incomplete information, they may not be aware of the profitability of innovation until motivated. Moreover, stringent regulation prevents firms from circumventing the intervention without improvement. As a result, firms with a competitive advantage based on their capacity to innovate will out compete those with a simple low-cost advantage (Porter and van der Linde, 1995).

Evidence from empirical studies suggests a less optimistic outlook, but it remains somewhat inconclusive. On the one hand, increased costs are expected to be associated with maintaining compliance, reducing the efficiency of capital invested while raising barriers to entry; on the other hand, regulation channels necessary investment into innovation and productivity in the long run (Koźluk and Zipperer, 2014). At the plant level, Greenstone (2001) compares manufacturing facilities in attainment and nonattainment counties following the 1970 Clean Air Act Amendment, which introduced the NAAQS. Drawing from microdata from the 1967–1987 Census of Manufacturers, he finds that nonattainment designation is connected to activity decline. During a 15-year period following the enactment of the amendment, roughly 590,000 jobs, $37 billion in capital stock, and $75 billion (in 1987 dollars) of output were lost in the manufacturing industry (Greenstone, 2001). However, on aggregate, the overall sector was only marginally affected. Productivity may also be diminished at least in the short run, as firms divert capital investment toward abatement rather than production. A follow-up study finds that nonattainment plants experienced productivity declines of up to 2.6 percent between 1972 and 1997, with ozone standards driving most of the negative effects and carbon monoxide standards associated with productivity gains (Greenstone, List, and Syverson, 2012). In this project, we consider Greenstone's work as an "anchor" paper for his rationale on the choice of treatment, but expand beyond the macroeconomic modelling he explores through employing causal methods using two-way fixed effects.

1.2 Research Questions

Most empirical work on environmental regulation focuses on outcomes at the plant, firm, or industry level, where regulatory exposure is easier to define and mechanisms are more directly observed. Much less is known about how these policies translate into aggregate local economic effects, in part because counties differ substantially in industrial composition, baseline pollution, and exposure to macroeconomic shocks. In this paper, we focus on investigating a county's overall economic activity when it becomes designated nonattainment according to the EPA Greenbook. Additionally we explore whether these effects are concentrated in sectors most plausibly exposed to compliance costs like manufacturing. We also ask whether the designation is associated with significant improvements in local air quality, which is central to the welfare rationale for regulation. We study these questions in the Rust Belt, where emissions intensive production is historically concentrated and the policy trade-offs between environmental quality and local output are especially salient.[1]

[1] Rust belt is defined as states Illinois, Indiana, Michigan, New York, Ohio, Pennsylvania, West Virginia and Wisconsin in alignment with Alder et al., (2014).

1.3 Geographic Context

The Rust Belt provides an interesting context to study the impact of environmental regulation, as the region is not only specialized in industrial activity but experienced a period of steep productivity decline post World War II until the 1980s. Between 1958 and 1985, the annualized weighted average of labour productivity growth in the Rust Belt was around 2.0% lagging the country by 0.6% (Alder et al., 2014). Root causes for this decline were attributed to labour unions fixing higher wage standards and a lack of competition in labour markets and output markets (Alder et al, 2014). Relative to other regions, lower competition and innovation led to output reduction, as the demand for the production of goods followed where innovation and efficiency were highest. It wasn't until a fall in minimum wage in the 1980s, which signified a restoration of competition in the labour market, that growth began to recover (Alder et al, 2014).

As a consequence of productivity decline, a reduction of manufacturing share in employment in the Rust Belt compared to the rest of the country during this period reflects a structural change in the industrial makeup. By centring the geographic focus of this project on the Rust Belt, we aim to examine whether environmental regulation in the region has contributed to further productivity enhancement and whether more recent post-2000 data offer additional insights into its industrial development.

2 Data

The EPA publishes annually nonattainment area and county information based on the Green Book for download on its website. In particular, the EPA Greenbook nonattainment list ("NAYRO") documents the years in which a county has received a nonattainment designation for each pollutant, from which we derive treatment status. Counties that have never received a nonattainment status are not included on the list; we define such counties as the "never-treated." Because the county-level, industry-specific output and income data are only available from 2001, our analysis will be restricted to the period of 2001 - 2010. The EPA also publishes daily air quality index (AQI) data at the county level for pollutants included in the Greenbook. We take the annual average of the data and use it as our outcome variable for pollution level. Economic outcomes data, including annual county-level, industry-specific gross domestic product (GDP), output index, and personal income, are obtained from the Bureau of Economic Analysis at the US Department of Commerce, spanning 2001 to 2010.

Regarding sample selection, we restrict the sample to counties in Rust Belt states that have never received a non-attainment designation, in addition to Wayne County, Michigan. This yields 556 never-treated counties, many of which have very small populations and output levels. Because our focus is on the effect of stringent environmental regulations on manufacturing hubs, such small counties are not relevant. We therefore define a major manufacturing hub as a county with both (i) an average population of at least 500,000 and (ii) an average manufacturing share of GDP above the sample median (approximately 16%) in the pre-treatment years of 2001 to 2004. The resulting samples include eight never-treated counties.

The control group consists of Marion County (IN), Hamilton County (OH), Montgomery County (PA), Monroe County (NY), Macomb County (MI), Lake County (IL), and Kent County (MI). By applying the sample selection criteria, this not only helps narrow down the sample size but also acts as controls for observable macroeconomic characteristics prior to applying our two-way fixed effects specification. As an overview of the sample during the pretreatment period, Table 1 displays the summary statistics of the county characteristics. We see that Wayne County has a relatively high population and output compared to the "never-treated" counties, reflecting Detroit's status as a major metropolitan area, while its per-capita income is comparatively lower. As a result, Wayne County lies in the tails of the distribution of sample counties along several dimensions. We discuss the implications of this feature for identification and estimation in the methodology section.

Table 1

Pre-treatment county characteristics (averages through 2004)

Variable	Wayne, MI	Control Mean	Control Median	Control 25th pct	Control 75th pct	Wayne Percentile
Average population	2,014,571	753,217	768,652	704,691	820,450	100.0%
Average GDP (all industries)	97,503,853	52,678,387	47,846,361	40,325,174	65,254,724	100.0%
Average per capita income	28,175	38,582	33,552	33,050	43,362	0.0%

We next examine the industrial profile of the treated county. As the Rust

Belt is historically characterized by a strong presence of heavy industries like manufacturing, we can verify this by examining Wayne County's sectoral decomposition (Table 2). Understanding the sectoral structure of Wayne County allows us to gain deeper insight into the industrial dynamics that define production in the treatment county. We use 2004, the year preceding treatment, as the benchmark for this analysis. While many broad industries can be further disaggregated into more granular subindustries (for example, durable goods within manufacturing, which we explore in later sections), we will focus on major industry groups. The classification of these main industry groups is based on county-level compensation data published by the Bureau of Economic Analysis.

Table 2
Industry composition of Wayne County, 2004

Rank	Industry	Value	Share
1	Manufacturing	15,657,241	15.77%
2	Government and government enterprises	12,983,148	13.08%
3	Real estate and rental and leasing	10,357,106	10.43%
4	Health care and social assistance	7,469,678	7.53%
5	Professional, scientific, and technical services	7,362,794	7.42%
6	Transportation and warehousing	5,420,165	5.46%
7	Retail trade	4,867,881	4.90%
8	Management of companies and enterprises	4,678,246	4.71%
9	Finance and insurance	4,536,677	4.57%
10	Construction	3,987,547	4.02%
11	Administrative and support and waste management and remediation services	3,115,922	3.14%
12	Accommodation and food services	2,922,445	2.94%
13	Other services (except government and government enterprises)	2,780,053	2.80%
14	Arts, entertainment, and recreation	2,118,123	2.13%
15	Information	2,098,041	2.11%
16	Educational services	1,012,512	1.02%
17	Mining, quarrying, and oil and gas extraction	131,130	0.13%

From this, we confirm manufacturing is indeed the most prominent industry in Wayne County, with government enterprises, real estate, health care, professional services, transportation and warehousing also being prominent industries.

The credibility of the two-way fixed effects (TWFE) design relies on the parallel trends assumption. As a preliminary check, we inspect the pre-designation trends by plotting the outcome paths for Wayne County alongside those for the “never-treated” counties to see whether they move similarly prior to treatment. If the pre-treatment trajectories show meaningful divergence, we instead rely on a synthetic control approach as a more credible alternative.

Figure 1 plots the evolution of PM2.5 and Ozone AQI in Wayne County and the ”never treated” counties. These are the only two pollutants with complete coverage for all counties in our sample over 2001–2010. Because our treatment of interest is Wayne County’s nonattainment designation for PM2.5, we focus the pollution analysis on this pollutant. Wayne County’s PM2.5 AQI follows the never-treated group fairly closely in the pre-treatment period, although there is a noticeable deviation in 2003 that raises concerns about the parallel trends assumption. We therefore use a synthetic control specification for this outcome.

Regarding output, Wayne County’s overall and manufacturing GDP appear more volatile than the “never treated” counties. In the top left panel of Figure 2, the manufacturing industry’s component in the GDP does not seem to be following the same path in the pre-treatment years between Wayne and the never-treated average. Given this result, we may not be able to draw causal conclusions using the TWFE set-up; the preferred specification for this outcome would utilize synthetic controls. In the right panel, the GDP of all industries in Wayne County and the never-treated county averages appear to follow a similar trend in the pre-treated periods. Wayne County's GDP saw a decline in the year following the nonattainment designation, while the never-treated counties remained on the original growth path. This suggests that the nonattainment designation may have negatively affected the county's total output value.

Additionally, we examine the quantity index in our analysis to check whether trends in output levels may be due to producing higher-quality goods in lower physical quantities. Increases in GDP may occur without changes in the quantity index, so the two measures may capture different margins of adjustment. The quantity index graphs show a similar story, suggesting analogous movements in line with GDP levels. There appears to be stronger evidence for the parallel trend assumption for the all-industries output level, but not manufacturing specifically. In the manufacturing sector, opposite movements in pre-treatment trends indicate possible violation of parallel trends. For the remainder of the analysis, given analogous movements, we will focus on GDP levels, which are a more conventional measure of economic activity.

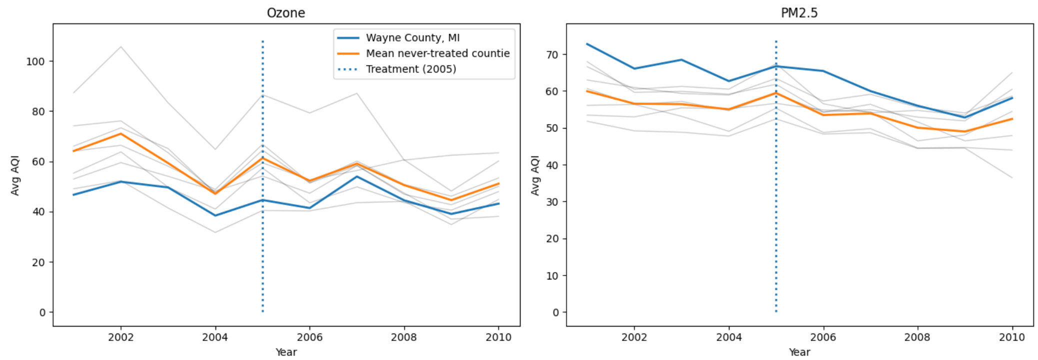

Figure 1
Wayne vs "never treated": pollution levels over time

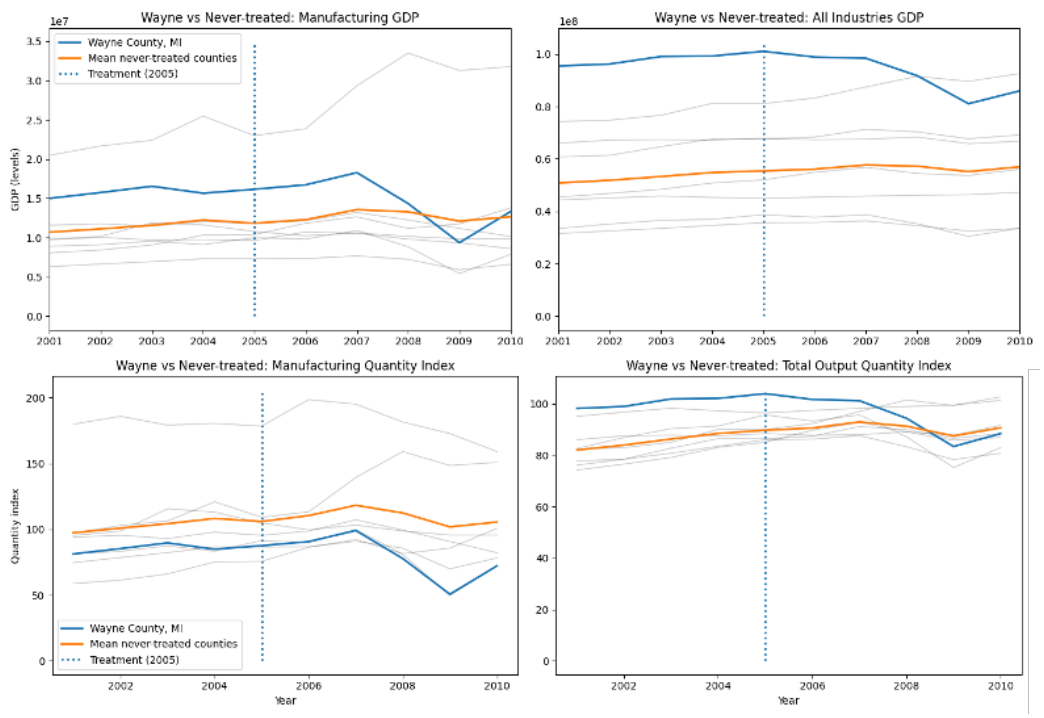

Figure 2
Wayne vs "never treated" counties: all-industries and manufacturing GDP over time

3 Empirical Strategy

Our choice of methodology is informed by the observations in the paper of Greenstone (2001), which documents systematic differences between attainment and nonattainment counties and examines the effects of the National Ambient Air Quality Standards (NAAQS) as a proxy for the stringency of environmental regulation. In particular, Greenstone (2001) shows that nonattainment counties tend to be more urbanized, have higher population density, higher average education levels, greater per capita income and government revenues, and are more likely to host carbon-monoxide–emitting facilities. These characteristics are persistent and difficult to fully control for using observable covariates alone.

3.1 Two-Way Fixed Effects

To address the concerns that nonattainment status is correlated with time-invariant observables and the broader macroeconomic conditions, we adopt a two-way fixed effects (TWFE) model as our main baseline specification to examine the dynamic effects of receiving a nonattainment status on the pollution and economic outcomes of Wayne County.

$$\begin{aligned} y_{it} = \alpha_i + \lambda_t + \sum_{k=2001}^{2003} \beta_k \, \mathbf{1}\{\tau_{it} = k\} \times \mathbf{1}\{\text{WAYNE}\} \\ + \sum_{k=2005}^{2010} \beta_k \, \mathbf{1}\{\tau_{it} = k\} \times \mathbf{1}\{\text{WAYNE}\} + \varepsilon_{it}. \end{aligned} \tag{1}$$

Equation (1) presents our baseline TWFE set-up. y_{it} is the outcome of interest in year t; α_i and λ_t are county and year fixed effects, respectively; $\mathbf{1}\{\tau_{it} = k\}$ is an indicator function for event year k, and $\mathbf{1}\{\text{WAYNE}\}$ is an indicator for the treated unit. The coefficient β_k is the parameter of interest, where β_k for $k \geq 2005$ represents the estimated effects in the post-treatment periods. The year prior to treatment, 2004, is normalized to zero as the reference period.

County fixed effects control for all time-invariant characteristics that differ regionally, while year fixed effects control for shocks and trends common to all counties in a given year. The identification comes from within-county variations based on the change in attainment status over time, relative to "never treated" counties, after netting out time common effects. Hence, the validity of the TWFE relies on two key assumptions: the parallel trends and no anticipation. In the absence of a non-attainment designation, Wayne County's outcomes would have evolved in parallel with those in the "never treated" group, and should not show evidence of divergence until the event year. Supporting evidence for these assumptions is obtained by assessing whether the pre-treatment coefficients in the TWFE specification are statistically indistinguishable from zero in the periods preceding treatment. Potential divergence in the paths of several of our outcome variables prior to treatment could suggest violation of key assumptions as discussed in Section 2 . For these outcomes, in addition of the TWFE model, we use synthetic control as an alternative approach.

3.2 Synthetic Control

The synthetic control is created with a weighted average of the never-treated group such that the difference in the pre-treatment levels between the never-treated and Wayne County is minimized.

$$\begin{aligned} \hat{w} = \arg\min_{w \in \mathbb{R}^J} \sum_{t=2001}^{2004} \left(Y_{1t} - \sum_{j=2}^{J+1} w_j \, Y_{jt} \right)^2 \\ \text{s.t.} \quad w_j \geq 0 \ \forall j, \qquad \sum_{j=2}^{J+1} w_j = 1. \end{aligned} \tag{2}$$

The weights (see Appendix) are obtained following Equation 2, where Y_{1t} is the outcome for Wayne County in year t and Y_{jt} is the outcome for county j in the never-treated group.

Using the synthetic control, we estimate the treatment effect in year t after the treatment as:

$$\begin{aligned} \hat{\tau}_t = Y_{1t} - \sum_{j=2}^{J+1} \hat{w}_j \, Y_{jt}, \\ \forall \, t \in \{2005, \ldots, 2010\} \end{aligned} \tag{3}$$

We treat the synthetic control as the counterfactual Wayne County. Therefore, identification relies on the assumption that a tight pre-treatment fit indicates the synthetic control approximates Wayne County's counterfactual post-treatment trajectory absent treatment. Conversely, observing a lack of fit in the pre-treatment period suggests that the synthetic control may not be a good proxy for Wayne County's counterfactual evolution, and thus a limitation in interpreting the results generated from implementing synthetic controls, which is especially relevant given the small set of "never treated" counties. [2]

4 Results

In this section, we will discuss the implications of receiving non-attainment status on each of the outcome variables of interest by applying our main TWFE specification and synthetic controls where relevant.

4.1 Pollution

We begin by investigating whether the impact of receiving non-attainment status has had a meaningful impact on emissions. We will focus on PM2.5 since this was the first nonattainment status Wayne County received during the period 2001 - 2010. Figure 3 presents the TWFE estimates of the dynamic effects of non-attainment status designation on Wayne County's PM2.5 AQI. As discussed in Section 2, the pre-treatment levels of PM2.5 AQI do not exhibit strong evidence for the parallel trend assumption. Consistent with this concern, two of the estimated lead coefficients are statistically significant at the 5% level, suggesting the presence of differential pre-trends. The point

[2] In this project, we compute synthetic controls weights using both custom-coded functions to demonstrate the mechanics as well as the scipi package for convenience.

estimates on post-treatment coefficients become negative starting in the second year. However, given the lack of evidence supporting the parallel trend assumption, these estimates are likely biased and requires further analysis to derive meaningful interpretations.

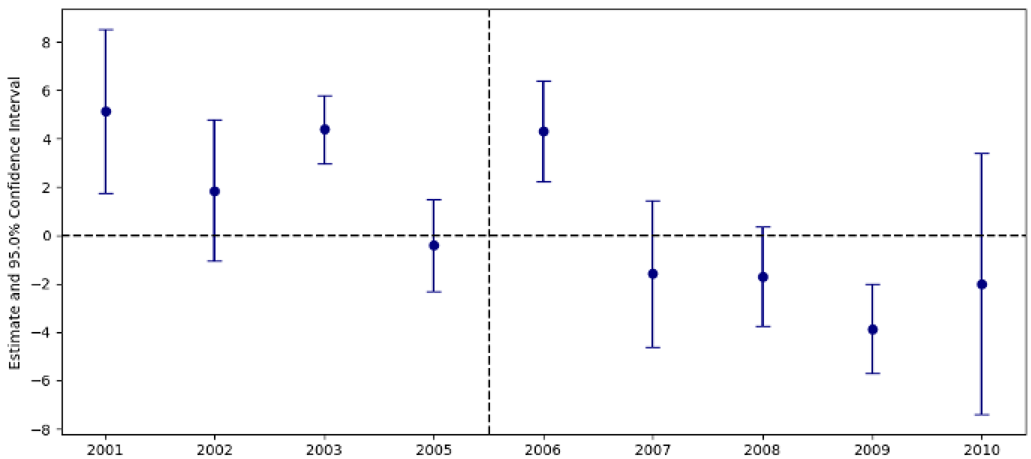

Figure 3
TWFE event study: the effect of nonattainment status on PM2.5 AQI

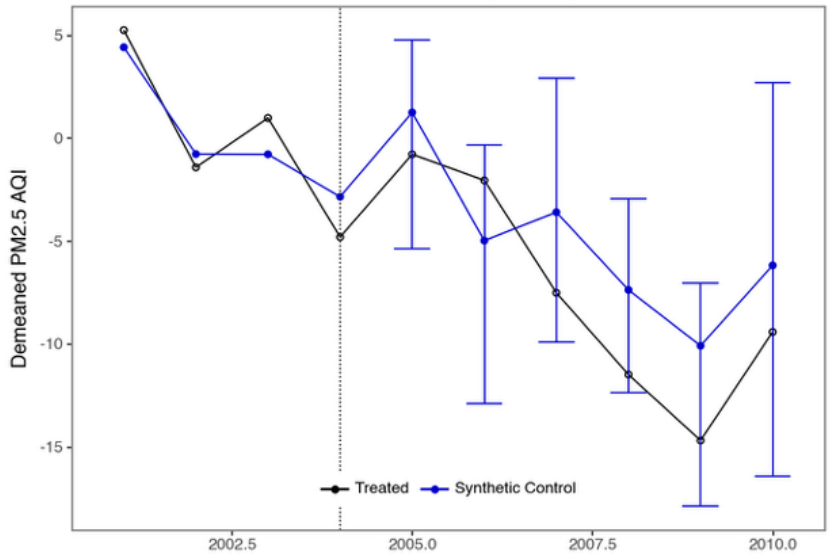

Figure 4
Synthetic control estimates of the effect of nonattainment status on PM2.5 AQI

With the likely violation of key assumptions, we then estimate the effect using synthetic control. To do this, we use demeaned AQI by subtracting the average pre-treatment level for each county as the outcome in the synthetic control setup, since Wayne County's PM2.5 emission is significantly above that of the "never treated" group. This methodology mitigates the issue of the weight optimization algorithm failing to generate weights that sum up to one.

As shown in Figure 4, in the pre-treatment period, the synthetic series broadly tracks the treated series. However, the fit is imperfect in several years, which cautions against over-interpreting small post-treatment deviations. Wayne County's demeaned PM2.5 AQI in the post-treatment periods became consistently below that of the synthetic control starting from 2007, which is mostly consistent with the TWFE results, but remains statistically insignificant. As such, from this alone, we can not conclude whether nonattainment status may have a meaningful impact on Wayne County's PM2.5 emissions.

4.2 Output

Next, we turn our analysis to the output outcomes of interest; we first examine the decline in overall output following Wayne County receiving the nonattainment status before exploring subindustries that may be driving the observed result.

4.2.1 All-Industries GDP

For all-industries GDP (Figure 5), the event-study estimates and confidence intervals in the pre-treatment period are centred around zero. This pattern is broadly consistent with the absence of strong differential pre-trends, providing suggestive evidence for the parallel trend assumption. In the first year of nonattainment designation (2005), the estimated effect is small but positive and statistically significant. In subsequent years, the point estimates turn negative, increase in magnitude, and are statistically significant. This provides evidence consistent with nonattainment status contributing to a decline over time. However, the pronounced decline observed in Wayne County relative to the control group beginning in 2008 may also reflect differential exposure to the financial crisis, rather than the effects of environmental regulation alone. Overall, this result shows evidence for the negative impact of the nonattainment status on the county's all-industries GDP. To understand the mechanism driving the decline, we now explore the effects of the policy on different components of the GDP.

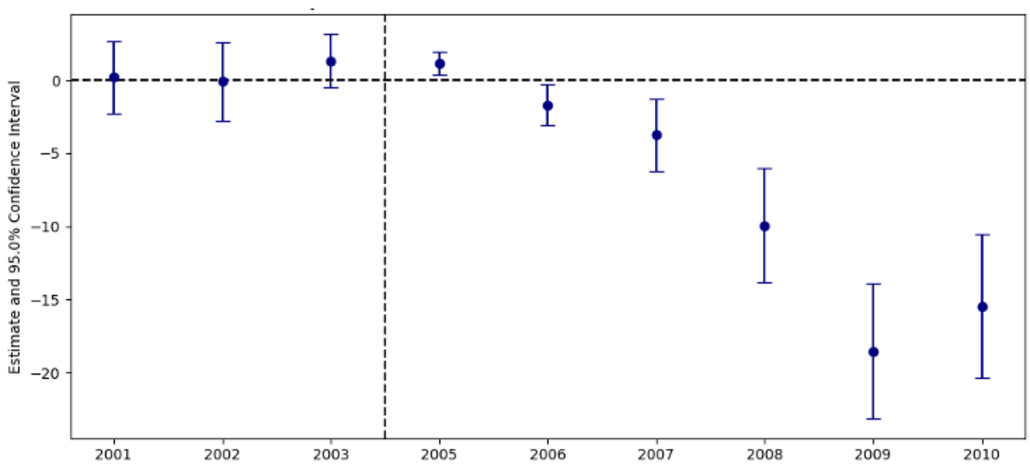

Figure 5
TWFE event study: the effect of nonattainment status on GDP (all Industries, millions)

4.2.2 Manufacturing GDP

As the manufacturing industry involves various heavy industrial processes, it is likely a major contributor to emissions and thus, more susceptible to stricter environmental regulations. This industry also accounts for the largest proportion of Wayne County's GDP, which we expect to have the most significant impact on overall output.

Figure 6 presents the TWFE estimates for the effect on the GDP of the manufacturing industry. The estimate is positive and significant in the second last period prior to the treatment, indicating possible violations of the parallel trend assumption. Although we observe negative coefficients for all-industries GDP starting from 2006, the estimates in the periods following the treatment remain positive but insignificant for the first three years and only turn negative in 2008. As discussed earlier, the results further from the event year should be interpreted with a greater degree of caution. Hence, these TWFE estimates do not suggest that the overall decline in GDP is driven by the manufacturing industry.

Nevertheless, it is also possible that nonattainment designation affects segments of the manufacturing sector differently (durable vs. non-durable goods manufacturing). For example, durable-goods manufacturing may rely on more pollution-intensive production processes than non-durable manufacturing, and is therefore more exposed to regulatory constraints and compliance costs. We check this by applying the same analysis to the durable goods manufacturing subindustry. As shown in Figure 7, the TWFE estimates for durable-goods manufacturing tell a similar story: the treated and control counties exhibit some pre-treatment divergence, and the post-treatment effects only turn negative in later years. The 2005 estimate is not available due to missing data.

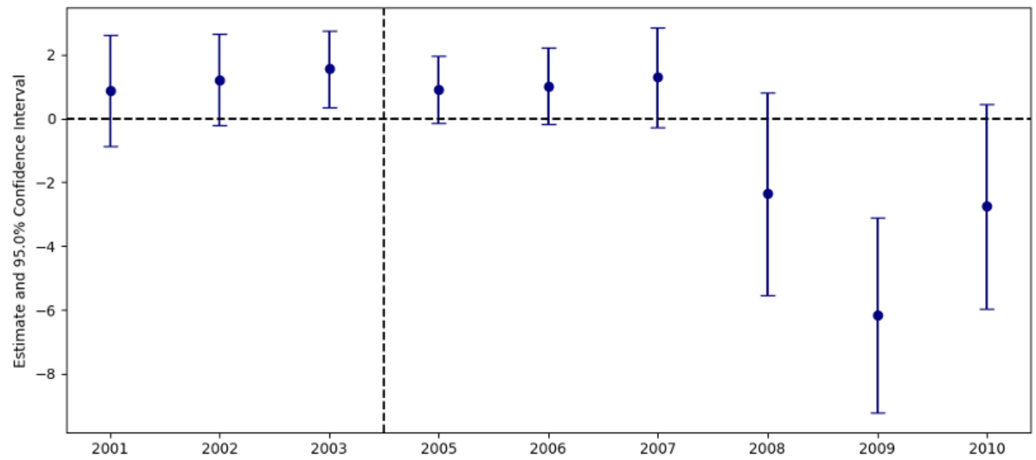

Figure 6
TWFE event study: effect of nonattainment status on GDP (manufacturing industries, millions)

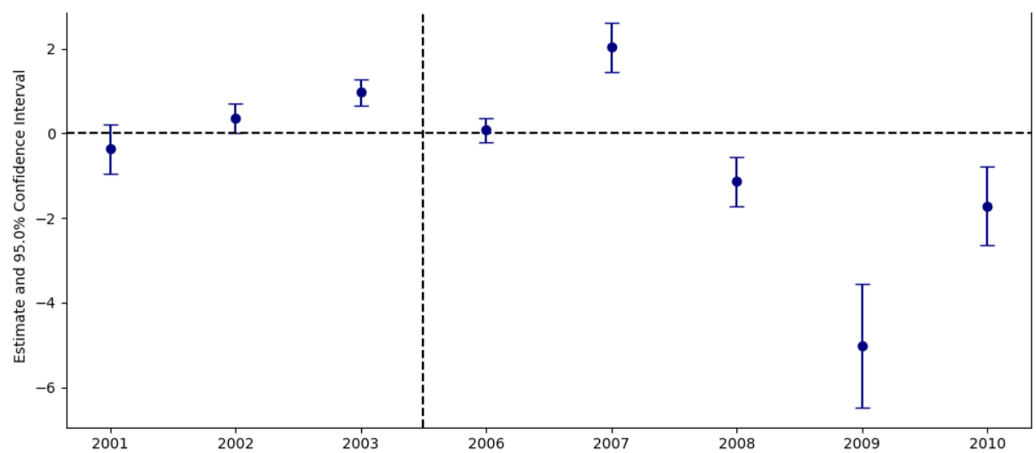

Figure 7
TWFE event study: effect of nonattainment status on GDP (durable goods manufacturing, millions)

To address possible violations of the parallel trend assumption from significant pre-trend coefficients, we then employ a synthetic control specification to estimate the effects on the GDP of manufacturing industries (overall and durable goods subindustry).

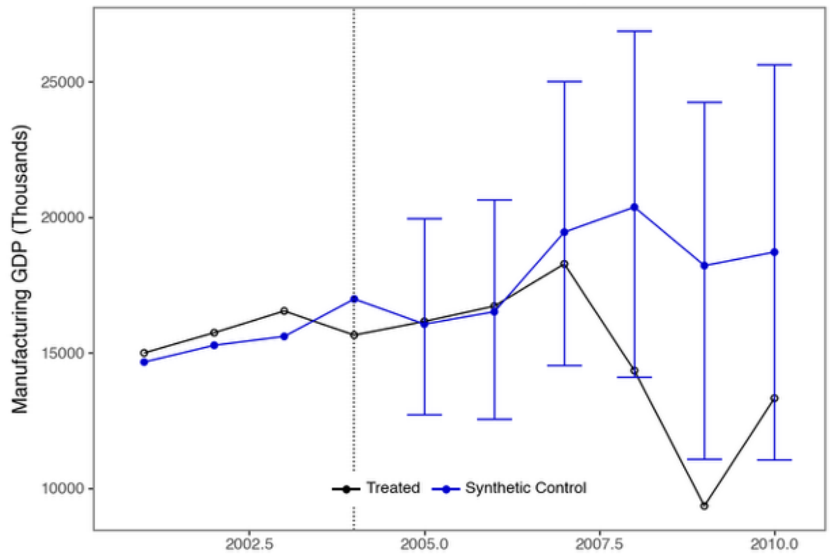

Figure 8
Synthetic control estimates of the effect of nonattainment status on manufacturing GDP (thousands)

For manufacturing output (Figure 8), Wayne County's pre-treatment level tracks the synthetic control closely in the first three years, but the gap widens in the year prior to treatment (2004). The estimates in the post-treatment periods are largely in line with the TWFE results, where we only observe significant, negative coefficients in the later years post-treatment, suggesting weak effects of receiving nonattainment status on Wayne County's overall manufacturing sector.

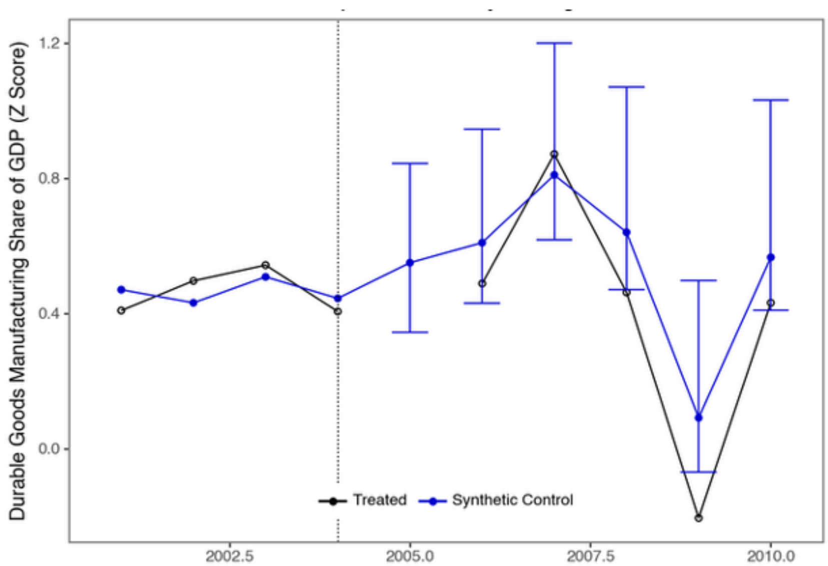

Figure 9
Synthetic control estimates of the effect of nonattainment status on standardized durable goods manufacturing share of GDP

Using GDP levels, we are unable to obtain synthetic control estimates for the effect of nonattainment status on durable goods manufacturing GDP because the optimization problem fails to converge. This issue can be mitigated by rescaling the outcome using a demeaned, standardized (z-score) measure of durable goods manufacturing's share of GDP. The results in Figure 9 show very similar pre-treatment levels across Wayne County and the synthetic control, suggesting adequate fit, but little significant deviation post-treatment, which is consistent with the overall industry.

In summary, neither the baseline TWFE results nor the synthetic control estimates show clear evidence of a decline in manufacturing, particularly in the early post-treatment years, indicating that manufacturing is likely not the driver for the negative impact on all-industries GDP.

4.3 Government Enterprises

Another GDP component we examine is output from government enterprises, which are publicly owned entities that provide market-like services such as utilities and transit systems. It is plausible that tighter emissions restrictions would affect these enterprises first. Relative to private firms, local governments have more direct authority over publicly owned operations and may be quicker to implement compliance measures, operational adjustments, or investment delays within them.

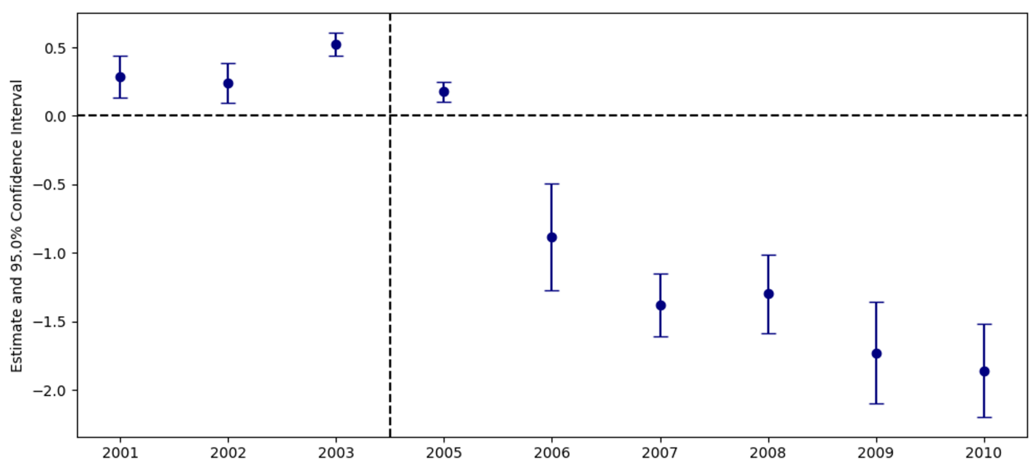

Figure 10
TWFE event study: effect of nonattainment status on GDP (government enterprises, millions)

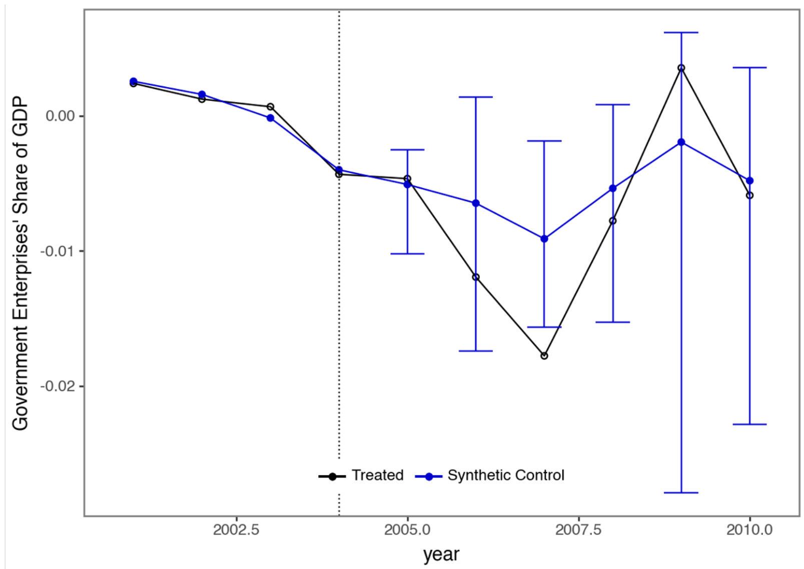

Figure 11
Synthetic control estimates of the effect of nonattainment status on government enterprises' share of GDP

The TWFE results (Figure 10) show significant, negative coefficients from 2006-2009, which correspond with the negative estimates we observe for all-industries GDP. However, since all the pre-treatment coefficients are also significantly different from zero, the concern for the validity of these estimates motivates us to employ the synthetic control design for this outcome.

For the synthetic control analysis, we use government enterprises' share of GDP rather than the level of government enterprise output, since the level specification exhibits convergence issues. In the pre-treatment period, the synthetic control tracks Wayne County closely, supporting the plausibility of the identifying assumption. In the post-treatment years, the estimated effects remain negative, but they are generally imprecise and no longer statistically distinguishable from zero, with the exception of 2007. Given the number of post-treatment periods examined, a single significant estimate should be interpreted cautiously due to multiple hypothesis testing concerns. Taken together, the synthetic control results do not provide strong, consistent evidence of a sustained decline in government enterprise output.

5 Discussion

Through a combination of TWFE and synthetic-control designs, we find limited evidence that nonattainment status affected PM2.5 emissions, but the estimates suggest a persistent decline in Wayne County's all-industries GDP following the designation. Evidence on the mechanisms behind this decline is largely inconclusive. In this section, we discuss limitations of our empirical designs, remaining threats to identification, and extensions for future work.

5.1 Policy Implications

Overall, the results suggest a potential trade-off. We do not find robust evidence that nonattainment designation reduces measured emissions over our sample period, but we do observe a persistent decline in all-industry GDP, indicating that the designation may impose meaningful economic costs. One interpretation is that compliance requirements raises costs or delay investment, even if these adjustments do not translate into large changes in county-level AQI measures or changes in the manufacturing industry specifically. From a policy perspective, this highlights the importance of pairing designation with effective implementation that directly address major emitters, while designing compliance pathways that minimize unnecessary disruption.

5.2 Limitations

5.2.1 Violation of the Parallel Trends and No Anticipation Assumptions

A key concern for identification is a violation of parallel trends and no anticipation assumptions driven by differences in growth trajectories. There is likely a positive correlation between production and pollution level. A county that expands output more aggressively than others in the pre-treatment period is both more polluting and more likely to receive a nonattainment designation. Moreover, this applies to industry dynamics as well; for example, if one county's manufacturing industry is growing faster than others prior to the treatment, then it is more likely to receive the treatment of a nonattainment designation. In this case, we have a selection into treatment due to a rising trajectory in manufacturing activities, conditional on the fixed effects.

This mechanism offers some explanation for why we observe Wayne County's pre-trends for manufacturing outputs and PM2.5 AQI above the control group. The autocorrelation in output level could also explain why the manufacturing GDP trajectory remains elevated in Wayne County three years after treatment. On the other hand, this selection mechanism is likely to have a smaller impact on all-industries GDP, which may be less tightly linked to the emissions thresholds that determine nonattainment status. We observe that for the aggregate GDP level, all pre-treatment period estimates are not significantly different from zero, which suggests that at least for aggregate output, we do not see strong evidence of systematic differences between Wayne County and the "never treated" counties leading up to the first nonattainment designation.

As a result, overall GDP exhibits more comparable and consistent pre-trends even when the manufacturing and components do not. However, even if this selection into treatment were a concern, it should have a limited impact on the validity of the negative coefficients we obtain from TWFE on all-industries GDP. In this case, this will likely bias the estimates towards zero, so the results obtained are a more conservative estimate.

5.2.2 Data

Data availability issue also weakens the empirical design; with only three pre-treatment periods, we have limited ability to assess whether Wayne County and the control group followed comparable trends over a longer horizon. In addition, the later-period estimates should be interpreted cautiously, since contemporaneous shocks, most notably the Great Recession, may have affected Wayne County differently than the control group in ways not fully absorbed by the fixed effects.

One possible alternative strategy as an extension of the project that may provide a more plausible identification assumption is a regression discontinuity design (RDD) that compares counties with emission levels just above or below the nonattainment threshold. Counties near the cutoff are likely to be similar in observed and unobserved determinants of economic outcomes. This approach could help address the concern that Wayne County's faster-growing manufacturing sector both raises pollution and increases the likelihood of treatment, which threatens parallel trends in TWFE.

5.2.3 Synthetic Control Pre-treatment Fit

The outcomes for Wayne County and the synthetic control often do not match perfectly in the pre-treatment periods, and the optimization problem for obtaining the synthetic weights sometimes fails without transformation of the data (e.g. standardization and demeaning). As discussed in Section 2, Wayne County is an outlier in many aggregate measures of our outcomes of interest, making it difficult to generate a synthetic control that matches its pre-treatment level with a limited number of counties in the "never treated" group. The identifying assumption requires that the synthetic control approximates Wayne County's counterfactual post-treatment trajectory absent treatment. One potential solution that would help improve the pre-treatment fit, and so provide stronger support for the identifying assumption, would be to have a larger pool of "never treated" counties. For future studies, one could expand the sample by including comparable counties from outside the Rust Belt region. When interpreting the results from the synthetic controls, it is important to keep in consideration that these limitations may dampen the strength of the results.

6 Conclusion

Focusing on Wayne County, Michigan, we examine how more stringent environmental regulations, as proxied by the nonattainment designation in the EPA Greenbook, affect outputs and pollution level. Overall, we find evidence of a persistent decline in Wayne County's all-industries GDP after designation, but little robust evidence that PM2.5 pollution (measured by county-level AQI) fell meaningfully over the same period.

Sector-level results do not clearly explain the aggregate downturn. Manufacturing and durable goods manufacturing show no consistent contraction in the early post-treatment years. Government enterprises display negative effects in the TWFE specification, but these estimates are weakened by pre-trend concerns and do not persist under the synthetic control approach. As a result, the main takeaway is an aggregate output decline with inconclusive evidence on both emissions reductions and specific industry mechanisms.

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Appendix

Table 3
Synthetic Control Donor Weights: Manufacturing Output

index	weight
Hamilton, OH	0.151852
Kent, MI	0.018608
Lake, IL	0.075756
Macomb, MI	0.084140
Marion, IN	0.439555
Monroe, NY	0.100872
Montgomery, PA	0.129217

Table 4
Synthetic Control Donor Weights: Manufacturing Output

index	weight
17097	0.0000000001
18097	0.4230716182
26081	0.0000000001
26099	0.2552308838
36055	0.0000000174
39061	0.3216974797
42091	0.0000000006

Table 5
Synthetic Control Donor Weights: Manufacturing share of GDP

index	weight
17097	0.0000000161
18097	0.0000000021
26081	0.0000000080
26099	0.0000000032
36055	0.0000000066
39061	0.0822792921
42091	0.9177206719

Table 6
Synthetic Control Donor Weights: PM2.5 AQI

index	weight
Hamilton	0.000000
Kent	-0.000000
Lake	0.392193
Macomb	0.000000
Marion	0.000000
Monroe	0.155015
Montgomery	0.452792

Table 7
Synthetic Control Donor Weights: Government Spending Share of GDP

index	weight
17097	0.0000036387
18097	0.0000486292
26081	0.9388926260
26099	0.0000026692
36055	0.0610383384
39061	0.0000079181
42091	0.0000061804

Table 8
Synthetic Control Donor Weights: Durable Goods Manufacturing Share of GDP

index	weight
17097	0.5317469377
26081	0.0000000038
26099	0.4682530306
36055	0.0000000020
39061	0.0000000017
42091	0.0000000242